

KBO STATS



Group 3

DATABASE MANAGEMENT 2

Final Presentation

2016311362 Sooyong Son
2020311081 Hyunwoo Kim

Contents

0. Intro

1. About Data

2. Application

3. Challenges

About Data



Data Scraping

Target : All available information of KBO players

- 2002~2021 regular season
- Records of defense(13995 rows) hitter(935 rows) pitcher(398 rows) runner(6212 rows)
- Records of player's salary(1700 rows)

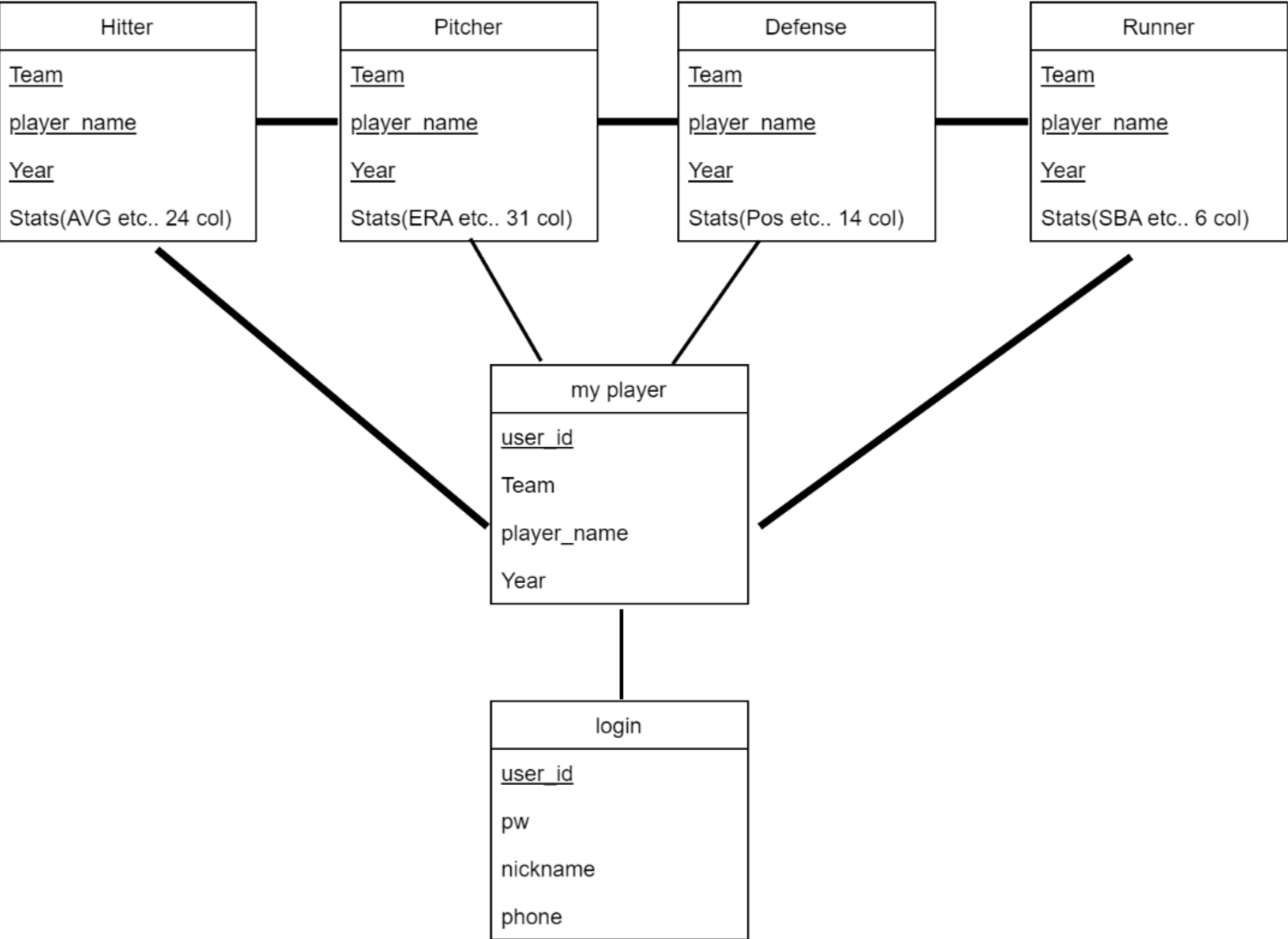
Sources

- KBO official site(<https://www.koreabaseball.com/>)
- Statiz(<http://www.statiz.co.kr/main.php>)

Tools



Data Structure



Tools



Team A took the top spot in the regular league with good momentum last year, but unfortunately missed out on the Korean Series.

You are the team leader.

Hot Stove League starts soon.



Application

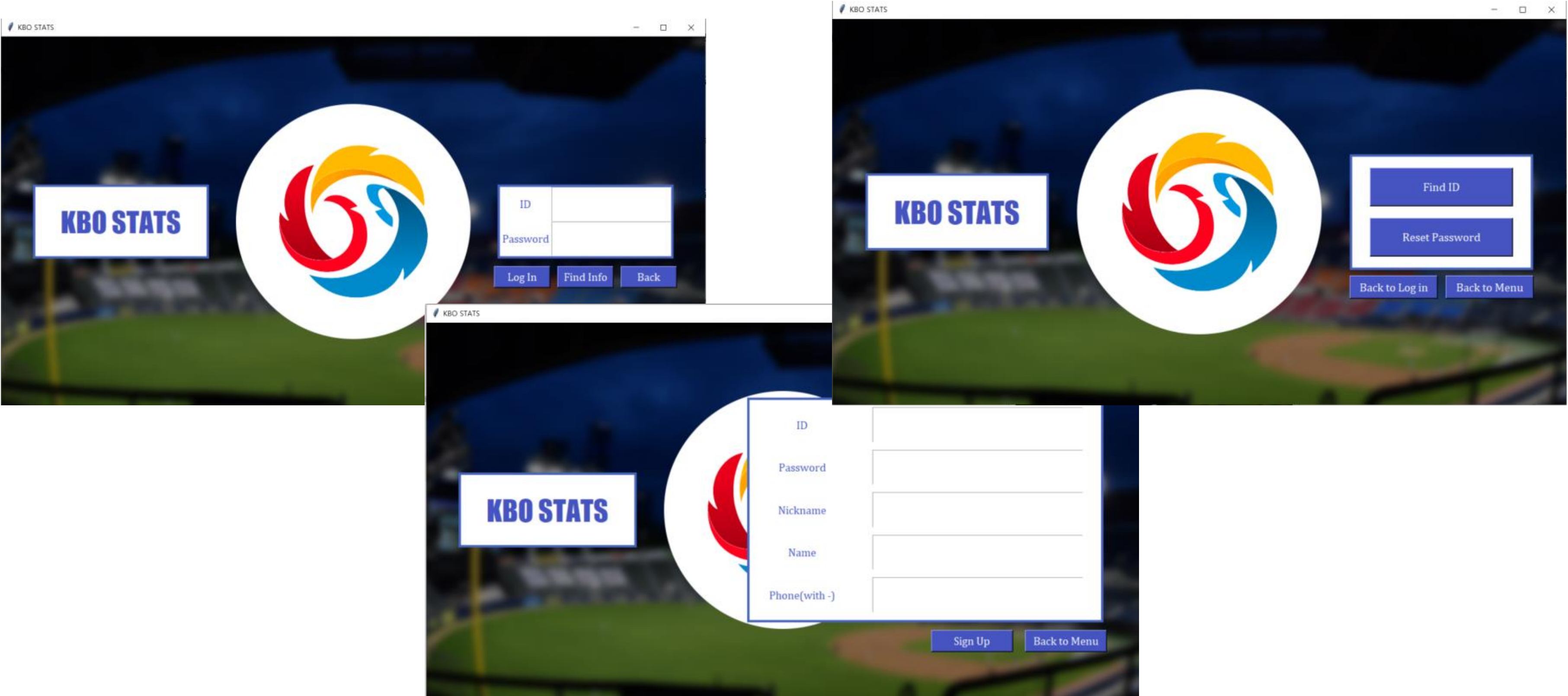


Situation 1.

You want to know the specific stats of some players you have watched during the season. you decided to try using the **KBO STATS** installed on the company intranet.

However, you are worried about whether you will have to **record** the stats of the players you have found. you also want to make sure that other team members, such as scout managers, **don't know which player you've been looking for.**

Log-in



The collage displays five different views of the KBO STATS website interface:

- Top Left:** A header section with a search bar containing "양익지", "2021", and "NC", and buttons for "Add My Player" and "Enter". Below it is a "Defense" table with columns: POS, G, GS, IP, E, PKO, PO, A, DP, FPCT, PB, SB, CS, CS%, and Year. The table contains one row of data for a player.
- Top Right:** A "Runner" table with columns: G, SBA, and a "Hitter" table with columns: AVG, G, PA, AB, R, H. The "Hitter" table contains one row of data.
- Middle Left:** A "Pitcher" table with columns: AVG, G, PA, AB, R, H. The table contains one row of data.
- Middle Right:** A "Manager" profile card showing a baseball bat icon, the nickname "Manager", and the name "sooyong" and phone number "010-3030-1010". Below the card is a "1" button and a "양익지" button.
- Bottom Left:** A "STAT" sidebar menu with options: COMPARE, SALARY, and MY PAGE.
- Bottom Right:** A "PROFILE IMAGE" selection grid with 20 icons representing different baseball equipment and player actions. Below the grid is a "Log Out" button.

Situation 2.

You have successfully logged in after registering as a member.

**All of a sudden, the ace player B of your team comes to mind
whose contract period ends this year.**

You start to think about how to find a good player to replace Player B.

Compare

KBO STATS

STAT

COMPARE

SALARY

MY PAGE

HitterPitcherDefenseRunner

2021KIAAVGAscendingEnter

0	선수명	Year	팀명	AVG	Select
1	김선빈	2021	KIA	0.307	Select
2	최원준	2021	KIA	0.295	Select
3	박찬호	2021	KIA	0.246	Select
4	터커	2021	KIA	0.237	Select
5	최형우	2021	KIA	0.233	Select

1/1

Compare

2021두산AVGAscendingEnter

0	선수명	Year	팀명	AVG	Select
1	박건우	2021	두산	0.325	Select
2	페르난데스	2021	두산	0.315	Select
3	허경민	2021	두산	0.278	Select
4	김재환	2021	두산	0.274	Select
5	양석환	2021	두산	0.273	Select

1/1

COMPARE

▼ 0.02	김선빈	선수명	박건우	
▲ 4	KIA	팀명	두산	▲ -0.02
▲ 39	0.307	AVG	0.325	
▲ 43	130	G	126	▼ -4
▼ 27	564	PA	525	▼ -39
▲ 5	501	AB	458	▼ -43
▲ 1	55	R	82	▲ -27
▼ 2	154	H	149	▼ -5
▼ 1	32	2B	31	▼ -1
▼ 1	0	3B	2	▲ -2
▼ 1	5	HR	6	▲ -1
▼ 1	201	TB	202	▲ -1
▲ 4	67	RBI	63	▼ -4
▲ 1	1	SAC	0	▼ -1
▼ 1	5	SF	6	▲ -1
-	2021	Year	2021	-
▲ 6	56	BB	50	▼ -6
-	2	IBB	2	-
▼ 10	1	HBP	11	▲ -10
▼ 33	39	SO	72	▲ -33
▲ 3	19	GDP	16	▼ -3
▼ 0.04	0.401	SLG	0.441	▲ -0.04
▼ 0.03	0.375	OBP	0.4	▲ -0.03
▼ 0.06	0.776	OPS	0.841	▲ -0.06
▲ 3	47	MH	44	▼ -3
▼ 0.01	0.32	RISP	0.325	▲ -0.01
▼ 0.25	0.0	PH-BA	0.25	▲ -0.25

Close

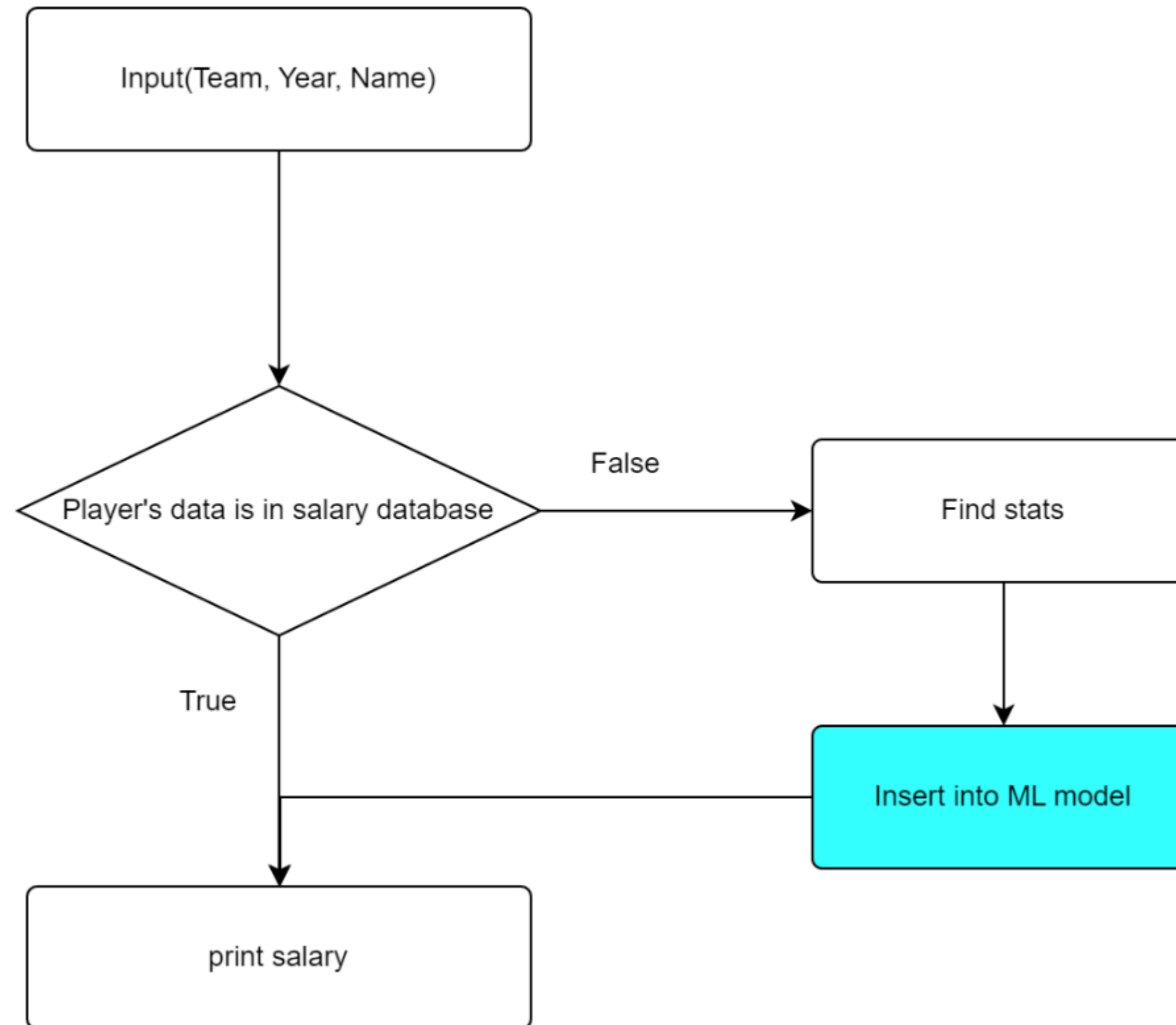
Situation 3.

You have found a good player C. However, an essential element in running a club is the **financial element**.

You want to know in advance the **appropriate salary level** for Player C, so that you have an advantage in the salary negotiation.

However, since player C has just made his professional debut, there is no salary information.

Salary



Salary

Machine Learning model

- * 2 Models

 - hitter model(41 features), pitcher model(39 features)

- * Standard scaled

- * Test with 4 models

 - LinearRegression, LinearRegression with regularization(Ridge, Lasso)

 - RandomForestRegressor

- * Find best model by comparing loss(RMSE). RFR(28831, 22935)

- * Tool



Salary

STAT

COMPARE

SALARY

MY PAGE

이종범

2002

KIA

Enter

Salary

43,000(existing)

Feature imporantnces of affecting salary(batter)

OOB	0.35
2B	0.08
BB	0.07
IBB	0.05
OPS	0.04
OBP	0.03
MH	0.03
GDP	0.02
R	0.02
SAC	0.02
SO	0.02
SLG	0.02
RISP	0.02
AVG	0.02
TB	0.01
PH-SA	0.01
SBA	0.01
WAR	0.01
AB	0.01

Feature impora

STAT

COMPARE

SALARY

MY PAGE

강백호

2021

KT

Enter

Salary

82000.0(predicted)

Feature imporantnces of affecting salary(batter)

OOB	0.35
2B	0.08
BB	0.07
IBB	0.05
OPS	0.04
OBP	0.03
MH	0.03
GDP	0.02
R	0.02
SAC	0.02
SO	0.02
SLG	0.02
RISP	0.02
AVG	0.02
TB	0.01
PH-SA	0.01
SBA	0.01
WAR	0.01
AB	0.01

Feature imporantnces of affecting salary(pitcher)

WHIP	0.25
AVG	0.25
TBF	0.07
SAC	0.04
SO	0.04
IP	0.03
OS	0.03
NP	0.03
H	0.03
2B	0.03
HR	0.03
HBP	0.03
3B	0.03
SF	0.03
WAR	0.03
R	0.03
BB	0.03
ERA	0.03
W	0.03

Challenges

- * There were lots of data, and it was difficult to think about how to effectively express it.
- * We couldn't find the player's id. It makes harder to merge or searching the data
- * Make machine learning model with proper steps like preprocessing, validation

THANK YOU:)